

Ordering Mixed Rational and Irrational Quantities

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|-------------------------------------|---|
| 1. 0.625 | 7. 3.14 |
| 2. 3.32 | 8. $<$ |
| 3. $<$ | 9. 1.4, $\sqrt{2}$, $\frac{3}{2}$ |
| 4. 4.25 | 10. $>$ |
| 5. $<$ | 11. $\frac{27}{5}$, $\sqrt{30}$, 5.6, $\sqrt{36}$ |
| 6. $\sqrt{5}$, $\frac{9}{4}$, 2.3 | 12. (a) the order = $\frac{15}{4}$, 3.9, $\sqrt{16}$, — |
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What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 13 tagged checks.

1. Write $\frac{5}{8}$ as a decimal.

Solution: Divide the numerator by the denominator. The division terminates after three places, which is what makes the number rational. 0.625

2. Give $\sqrt{11}$ to the nearest hundredth.

Solution: $9 < 11 < 16$ brackets the root between 3 and 4. Squeezing: $3.31^2 = 10.9561$ and $3.32^2 = 11.0224$ trap the radicand, and the upper one is nearer. 3.32

3. Write $<$, $>$, or $=$ between $\frac{7}{9}$ and 0.78.

Solution: The fraction repeats: $7 \div 9 = 0.7777\dots$. The two decimals agree in the tenths place and first differ in the hundredths, where seven is less than eight. $<$

4. Write $\frac{17}{4}$ as a decimal.

Solution: Four goes into seventeen four times with one left over, and the remainder makes a quarter. 4.25

5. Write $<$, $>$, or $=$ between $\sqrt{17}$ and 4.2.

Solution: Square the decimal instead of squeezing the root, which is faster here: $4.2^2 = 17.64$, which is larger than the radicand, so the decimal is larger than the root. $\boxed{<}$

6. Order $\sqrt{5}$, 2.3, $\frac{9}{4}$ from least to greatest.

Solution: Convert all three to two decimal places: $\sqrt{5} \approx 2.24$ (since $2.23^2 = 4.9729$ and $2.24^2 = 5.0176$), $\frac{9}{4} = 2.25$ exactly, and 2.3 is already a decimal. Ordering the decimals and rewriting in the original forms:

$$\boxed{\sqrt{5} < \frac{9}{4} < 2.3}$$

7. Give π to the nearest hundredth.

Solution: The decimal expansion of the circle constant begins 3.14159..., and the digit after the hundredths place is below five, so it rounds down. $\boxed{3.14}$

8. Write $<$, $>$, or $=$ between π and $\sqrt{10}$.

Solution: To two places both read 3.14 and 3.16; to three, $\pi = 3.142$ while $\sqrt{10} = 3.162$ (check: $3.16^2 = 9.9856$, just under the radicand). The root is the larger. $\boxed{<}$

9. Order $\sqrt{2}$, $\frac{3}{2}$, 1.4 from least to greatest.

Solution: $\sqrt{2} \approx 1.41$ (since $1.41^2 = 1.9881$ and $1.42^2 = 2.0164$), $\frac{3}{2} = 1.5$ exactly, and 1.4 is already a decimal. The plain decimal is smallest, and the fraction is largest.

$$\boxed{1.4 < \sqrt{2} < \frac{3}{2}}$$

10. Write $<$, $>$, or $=$ between $-\sqrt{9}$ and -3.1 .

Solution: The radicand is a perfect square, so the first quantity is exactly -3 . On the number line -3.1 sits further left, and further left means smaller, so the first quantity is the larger. $\boxed{>}$

11. Order $\sqrt{30}$, 5.6, $\frac{27}{5}$, $\sqrt{36}$ from least to greatest.

Solution: $\sqrt{36} = 6$ exactly, $\frac{27}{5} = 5.4$ exactly, $\sqrt{30} \approx 5.48$ (since $5.47^2 = 29.9209$ and $5.48^2 = 30.0304$), and 5.6 is already a decimal. Note that the largest of the four is a root, not the plain decimal.

$$\boxed{\frac{27}{5} < \sqrt{30} < 5.6 < \sqrt{36}}$$

12. **Challenge.** Nadia orders $\sqrt{16}$, 3.9, $\frac{15}{4}$ by reading the printed numerals. (a) Correct order? (b) Why does that not work?

Solution (a): Convert first: $\sqrt{16} = 4$ exactly, $\frac{15}{4} = 3.75$ exactly, and 3.9 is already a decimal.

$$\boxed{\frac{15}{4} < 3.9 < \sqrt{16}}$$

Solution (b) — instructor-judged. Nadia compared the numerals as printed rather than the quantities they stand for. A radical and a fraction give no reliable size information until they are evaluated: the root here is not even irrational, and it turns out to be the *largest* of the three despite showing the biggest numeral inside it. A full-credit answer names the conversion step as the thing that was skipped.